10.7 Circles in the Coordinate Plane

Name _____ Period ____ Date _____

Geometry 10.7 (30-minute plan) · Student sheet

I can…

- write the equation of a circle from its center and radius.
- find the center and radius of a circle from its equation and graph it.
- decide whether a point is on, inside, or outside a circle.

Vocabulary · fill in the missing word

Standard equation of a circle — $(x - h)^2 + (y - k)^2 = r^2$, where (h, k) is the _____ and r is the radius

Center — the point inside a circle that is the same _____ from every point on the circle

Radius — a _____ from the center to a point on the circle (or its length)

Part 1 · Notes and examples (fill in each note, then work the example)

Standard equation of a circle: $(x - h)^2 + (y - k)^2 = r^2$, with center _____ and radius _____. With the center at the origin: $x^2 + y^2 = r^2$.

Example 1

Write the standard equation of each circle.

a. center $(-2, 5)$, radius 3

b. center $(0, 0)$, radius 4

From the center and a point: find the radius with the _____ formula, then write the equation.

Example 2

A circle has center $(1, -3)$ and passes through $(4, 1)$. Write its standard equation.

Not in standard form: _____ for x and for y, then read the center and radius.

On, inside, or outside: compare the point's distance from the center with the _____.

Example 3

Find the center and radius of the circle $x^2 + y^2 - 6x + 4y - 3 = 0$.

Part 2 · Partner practice

Partner practice

Find the center and radius of the circle $x^2 + y^2 + 8x - 2y + 8 = 0$.